

Chapter 9

Evaluating the Patient

The typical orthopedic exam determines the range of spine motion, some neurological measures such as the strength of reflexes, and perhaps some qualitative measures of muscle strength. These measures provide little guidance for designing prevention and rehabilitation programs. Further, we published a study (Parks et al., 2003) in which we tracked back pain patients in a pain clinic. Scores obtained from patients' assessment had very little relation to who recovered and returned to work. Quite often, these numbers are more for the legal determination of disability than for aiding in the clinical decision process. For example, some erroneously believe that all patients should have normal to average range of motion (ROM) values even though they probably were not average prior to becoming a patient. The tests discussed in this chapter offer more useful indicators of pathology to assist you in making treatment decisions.

Of the several deficits in low back variables identified in earlier chapters, many are the direct result of injury. They include aberrant lumbar motion patterns, perturbed motor patterns of muscle recruitment, and aberrant joint motion with concomitant pain and loss of muscle endurance. Unfortunately, testing for these deficits is not easy. The challenge is to find the tests that can best identify the deficits and that are reasonably safe and do not require expensive or specialized equipment. The tests that come closest to meeting these criteria are described in this chapter, along with guidelines for tests to quantify patient deficits. The results form the rehabilitation objectives for the patient and provide clues for designing exercise. This chapter also discusses how patients can help to define their own rehabilitation targets.

There is no such thing as nonspecific back pain—there are only people who have not had thorough assessments. Provocative testing, when combined with movement screens for joint symmetry of motion, strength, and endurance, underpins a powerful classification for people with back pain. Classification enhances the therapy plan and identifies what to avoid. The process continues throughout the recovery process to define tolerable levels of load in specific postures and movements so that the dosage of therapeutic

exercise can be tuned to the person.

The general approach begins with an interview followed by an assessment of movements, postures, and loads that cause pain. Then the patient is shown the cause of the pain. This is crucial for learning to avoid the cause through movement technique. Then, and only then, is corrective and therapeutic exercise considered. Yet too many clinicians unwittingly engage the patient in strengthening routines without this process, and fail to achieve a pain-free state.

Most Crucial Element in Evaluation

Before learning about any tests or techniques, it is vital that you understand the central element in all diagnoses: your brain! If your own observation and reasoning skills are not well developed, the best tests and the most advanced technology will be of little use, as the following text will show.

Geoff Maitland, the well-known Australian physical therapist, promoted the hypothesis formulation approach to diagnosis—which is very similar to our own. Like a crime detective, a clinician must consider evidence from all sources. As each piece is considered, the hypothesis is either strengthened or weakened. However, unlike the detective, the expert clinician can obtain a definite history, even though a patient may present conflicting signs and disguised characteristics. This complexity means no more than that the patient has a complex presentation. Such patients may be farther down the degenerative cascade of tissue and nervous system change, or they may have some biological processes under way that complicate the search. Biomechanical knowledge is critical for success in hypothesis formation. Usually, the overused tissues are the ones involved in symptom creation. The solution often lies in changing the biomechanics to avoid loading painful tissues.

A keen clinical eye is a feature of all great clinicians. I often tell the story of when I was invited to teach at a renowned spine center and part of my course was to conduct three examinations in front of the center's 18 clinicians. The clinicians included orthopedists, neurologists, physical therapists, and so on. We went to the waiting room and were introduced to the first patient. The patient rose from the chair and walked to the large training room that served as our exam room. Then I asked the clinicians to turn away from the patient and face the wall. I asked them to tell me about the patient—and not one of them could. I told them that they had all failed.

Interpreting Patient Presentation

Clinical Relevance

Form a working hypothesis and continually reassess the hypothesis, doing

the following:

1. Observe everything, starting with the person's sitting posture, how she rises from the waiting room chair, how she stands, and how she walks.
2. Elicit and record the history: link injury mechanisms and pain mechanisms.
3. Perform provocative tests to determine the loads, postures, and motions that exacerbate the pain and those that relieve it.
4. Perform functional screens and tests to determine whether perturbed postural, motion, and motor patterns exist.

I then described all that I had observed, beginning with the seated posture of the patient, which was a full-flexion, slumped posture (this usually suggests flexion intolerance). Rising from the chair was initiated with more spine flexion and less hip extension. Then the hips rose, indicating typical hamstring dominance for hip extension and gluteal deficiency in generating the hip extensor torque. The spine extended last with spine extensor muscle activation. This strengthened the working diagnostic hypothesis that was forming in my head. The classic antalgic walking posture was shown with arms swinging and bending at the elbows instead of the shoulders. A keen ear could hear the asymmetry in footfall with the right-sided deficit in tibialis anterior. Then I observed the classic standing posture with the weight shifted predominantly onto one leg.

After pointing out these features, I stated that my working diagnostic hypothesis was a disc bulge on the right posterolateral side of L4-L5. Further testing showed this to be correct. The other two patients included a stenotic person and one with a highly unstable spine lumbar hinge with over stiffness in the thoracic region. After conducting the same qualitative observations on these two, the clinicians understood what I meant when I said that their eyes, ears, and hands were their best scanners. Interestingly, a few older clinicians stated that they had forgotten their observational skills because of access to high-tech medical imaging—this was a poignant reminder to rehone their observation and assessment skills.

What Every Clinician Must Know

Assessing a patient usually reveals several dysfunctions. I have seen patients with a litany of dysfunctions in the referral, all with prescribed treatment plans. Yet none of the dysfunctions listed caused their back pain. As clinicians, we need to focus on reducing the pain first, meaning address the big issues and show the patient quickly how to reduce the pain. Teaching him how to activate the transverse abdominis rather than showing him how to get out of a bed pain free is a disservice. The painless dysfunctions can be addressed later.

The implication of the previous paragraph is that the clinician must have expert competence in, skill in, and knowledge of the following:

- Injury and pain mechanisms as well as mechanisms of pain desensitization and tissue repair. Those without this knowledge make statements such as, We do not know the source of pain. Those with this knowledge and skill use the precise application of stress to a targeted structure, or tissue within a structure, while looking for the onset of painful symptoms. They can replicate the patient's familiar pain or create new pain.
- The information content of each test and an understanding of what each test reveals. Some simple tests are reliable but limited in what they reveal. The more involved tests are not reliable and require clinical skill, but have great potential to reveal the pain mechanism. Both are necessary to understand patients' pain.
- How to adjust each provocation test posture, load, range, trajectory, and neural and other tissue tensions to deeply probe the mechanism of the pain.
- Corrective and therapeutic exercise selection, dose, technique modification, and progression.
- Comorbidities that cause or mask back pain. Sembrano and Polly (2008) diagnosed 289 patients presenting at a spine surgeon's practice. Of these, 65% had spine-only pathology, and 17.5% had additional hip or sacroiliac (SI) pain sources. Eight percent had hip SI joint pathology (or both) with no spine pathology, and 10% had an undefined pain source. This is consistent with our clinical experience.
- Subgroups of patients. Average clinicians may have three or four subgroups that are assigned specific treatments or approaches. Highly skilled clinicians have finer criteria to the point that each patient

becomes a unique case study.

- Variables that influence the assessment. A patient who is flexion posture intolerant will experience exacerbated pain symptoms from the drive to the assessment appointment. The keen clinician understands that her observations are modulated by the prior loading history of the patient.
- The influence of perception and personality on the presentation.

What Every Patient or Client Needs to Know

The medical system does not always give all involved parties the necessary information to optimize recovery. All workers with back pain need to know the following to facilitate their recovery:

- Exam results. Their current scores give context to their goals.
- Natural history and prognosis. There is no evidence that back disorders last into retirement; in fact, they are often addressed with appropriate classification and treatment plans.
- Causes of pain. Patients are often amazed to learn that the way they move and activate muscles can eliminate pain.
- What to avoid. Removing the cause of the disorders is an obvious strategy; this also improves the efficacy of the therapy.
- Recovery plan. A progression should begin by addressing the movement disorders with corrective and therapeutic exercise, and then move on to stabilizing body areas needing stabilization and mobilizing those that need mobilization. Next in the progression is enhancing endurance so that joint-sparing movement patterns can be repeated even when fatigued, followed by building some strength and possibly some power-generating ability at the hips and shoulders if appropriate.

First Clinician–Patient Meeting

Every person with back pain has limited capacity for physical work. Every aspect of therapeutic exercise must be justified; otherwise, capacity is wasted. The objective is to determine the capacity, determine what is tolerable, determine the deficits, and design the best therapy to rectify the shortcomings. Having a road map to follow will allow you to plan for your patient the fastest and most efficient journey to optimal back health. Finally, you should proceed only after you have screened your patient for all red flag conditions. The following checklist will help you determine appropriate rehabilitation exercise.

1. Identify the rehabilitation objectives (specific health or performance objectives). The specific rehabilitation objective determines the acceptable risk-to-benefit ratio. A performance objective carries higher risk. Because the principles of bodybuilding and athletic training are so pervasive, you need to be sure that all patients understand the difference between athletic performance objectives and those for pain reduction and improved daily function. This implies that the recovery plan should first direct all efforts toward pain elimination and then when successful, address enhancing athleticism.
2. Ask key questions. Ask the patient, Do your symptoms change in intensity? Do you have better and worse periods? If the answer is yes, you are guaranteed success. Identify what makes the pain worse and what alleviates it. Ask, Do you get a sharp pain rolling over in bed? If the answer is yes, the patient has instability to the point that when the spine is not controlled with muscle stiffness, joint micromovements cause sharp pain. He will most likely do well learning to replicate these relieving patterns. Another key question is Do you have morning stiffness or pain that diminishes throughout the morning, or does the pain ramp up throughout the day? Pain that ramps up usually indicates that loading causes cumulative symptoms. Teaching the person “pristine” spine-sparing movement strategies will delay the symptom onset. Morning stiffness may indicate an inappropriate bed or mattress. Follow up this suspicion by asking whether the person has a different pain pattern after sleeping, for example, in a hotel bed. Other questions

and responses hone the focus for the remainder of the interview stage.

3. Consider patient's age and general condition. Younger patients (from the teens to the fifth decade) tend to have more discogenic troubles, whereas arthritic spines tend to begin developing after the age of 45, and stenotic conditions develop after that. Note how patients walk and sit. Are they in noticeably poor condition, either emaciated with little muscle mass or heavy with fat rather than muscle? It is also assumed that patients have been medically screened and cleared for cardiovascular concerns, or any red flag conditions such as tumors.
4. Identify occupation and lifestyle details. Generally, you should begin by documenting patients' daily routines: when and how they rise from and retire to bed, meal routines, and exercise and recreational habits. Then direct specific focus toward areas of concern. For example, if the patient reports watching TV for 2 hours in the evening, ask for details on the type of chair, range of postures used, and so on. After gathering information about the patient's daily routines, inquire about occupational demands. All of this information, when added to the clinical presentation, will help you evaluate common links. Discogenic troubles are linked with prolonged sitting (particularly prolonged driving) and repeated torso flexion. A passive or inactive lifestyle is also associated with disc troubles. Arthritic conditions, facet troubles, and the like, are more linked with jobs and activities that involve large ranges of motion and higher loading. Former athletes such as soccer players also fall into this category, although long-distance runners do not because they do not, presumably, take the spine to the end ROM.
5. Consider the mechanism of injury. Attempts to re-create injury mechanisms are fruitful only when the real mechanisms are understood. These were detailed in chapters 3 and 4. Once identified, the mechanisms can be linked with specific tissue damage (much of which is otherwise not diagnosable). Not only will this assist in designing the therapeutic exercise, but it will also help in teaching patients to avoid loading scenarios that could exacerbate the damage and symptoms. Note that some of these injury mechanisms will have acute onset, whereas others progress slowly. Slow onset may result in some patients' being unable to identify the mechanism of injury. Nevertheless, a culminating event is usually involved. Careful questioning about events leading up to that event will provide clues as to the mechanisms of injury.

6. Have the patient describe the perceived exacerbators of pain and symptoms. Prompt the patient to describe the tasks, postures, and movements that exacerbate the pain. Examine these tasks from a biomechanical perspective to determine which tissues are loaded or irritated. These tissues should be spared in the exercise therapy, and the exacerbating movements minimized with movement pattern coaching.
7. Have the patient describe the type of pain, its location, whether it is radiating, and specific dermatomes and myotomes. Descriptions of the type of pain are usually helpful; patients may describe their pain as deep and boring, scratchy, sizzling, at a particular point, general over the back region, continually changing, and so on. You may need to help some people describe their pain by offering adjectives to choose from. In chapters 3 and 4 I described the link between pain types or patterns and specific tissues and syndromes. Keep in mind that changing symptoms over the short time of an examination generally suggest more fibromyalgic syndromes, which can sometimes be resistive to exercise therapies—particularly ones that cause pain. The issue here is that pain-free motion must be found and repeated, so that people can slowly expand their repertoires of pain-free motion.
8. Consider patient personality and perception. Patients perceive similar symptoms differently. For example, the patient who exhibits self-manipulating behavior during the interview (e.g., twisting the spine to initiate a crack) usually give high priority to the stretch reflex. The same patient will likely report enjoying yoga and daily stretching. In contrast, another patient perceives stretching as discomfort—she interprets the stretch reflex differently. These two patients require different clinical approaches. In a similar fashion, some perceive muscle work as pain, whereas others perceive the difference between muscle work and back pain. Distinguishing these two presentations, and coaching patients in interpreting their perceptions, will influence their compliance. The influence of personality on compliance is poignant when we work with experimental groups of diabetics and breast cancer survivors at our university clinic. The buzz in the room with each group is the polar opposite of that of the other. When the cancer survivors are asked to perform 10 repetitions of a therapeutic exercise, typically they perform more when the clinician is not looking. They are typically type A people, a group of overachievers who take control to regain their fitness.

The diabetic group is typically the opposite, being made up of more type B personalities. When those with diabetes are asked to do the same 10 reps, they bargain for why they need to do only 3. They are generally, as a group, movement averse. Personality influences their presentation and response to advice. I consciously note whether the patient will need encouraging to engage in therapeutic exercise or will need holding back so he does not work until pain erupts.

9. Take dermatomes and myotomes into account. With radiating symptoms, the dermatomes and myotomes can assist in understanding the involved segmental levels and whether the pain originates from a specific nerve root. For example, direct pressure on the root could indicate a unilateral disc bulge or end-plate fracture that would cause a loss in disc height together with a loss of root outlet foramen size. In this way the spinal level can be linked with the dermatome or myotome but not the actual tissue damage. Further, nerve root pressure can occur at a specific spinal level on the outlet nerve, consistent with a dermatome or myotome, or on the traversing nerve from above if there is pressure on the cauda equina centrally. Thus, dermatomes and myotomes are another consideration when forming an opinion from the results obtained from several tests that can include medical imaging and provocative tests.
10. Perform provocative tests. Having observed the patient sit, rise from a chair, stand, and walk, you have developed a working diagnostic hypothesis. Once you suspect that specific tissues are damaged or sensitive, you can load them to see whether loading produces pain. This is provocative testing. Many patients have more complex presentations, with several tissues involved. Nonetheless, the provocative procedure still indicates which postures, motions, and loads cause pain and specific motions, postures, and loads that should be avoided when designing the therapeutic exercise. Generally, patients' descriptions of the activities they find exacerbating of their pain (item 6 on this list) will guide your decision as to which tissues to load and stress. For example, lumbar extension with a twist can provoke the facets, whereas the anterior shear test may be warranted for suspected instability (shown in the Testing for Lumbar Joint Shear Stability).
11. Perform functional screens. To determine whether a patient is moving in a spine-conserving and efficient way, use functional screens. In addition, functional screens indicate the suitability of a specific exercise or serve

as qualifying tests prior to exercise prescription.

Interpreting Pain Reports Obtained During the Interview

Clinical Relevance

A skilled clinician can often create a very robust hypothesis simply by listening to the patient's story and asking strategic questions. This sidebar presents observations from my own clinical experience given the fact that we follow up with every patient who comes to our research clinic. This allowed us to assess how effectively we interpreted patients' pain reports and how well they responded to our specific recommendations.

- Of course, recognizing red flag conditions is essential. These include a history of trauma, bowel or bladder incontinence, a history of cancer, unexplained weight loss, intravenous drug use, systemic illness, low back pain with fever, progressive and constant night pain, abdominal pain (between the navel and pubis), saddle anesthesia, and no response to motions, postures, and loads.
- Disc symptoms that are radiating generally centralize, or are relieved with specific postures and movements. Onset can be sudden. A feeling of giving way and other similar descriptions of instability in neutral joint ranges are common. Interestingly, reported dermatome distributions are more accurate than neurological signs. Although radicular patterns (pain, dermatomes, myotomes) demonstrate anatomic variability, the following are most typical:
 - L1, L2: Pain in the lower abdomen, groin, and medial thigh with hip flexor myotomes
 - L3: Pain in the anterior thigh and knee, with knee extensor myotomes
 - L4: Pain in the medial malleolus and small toes, with an ankle dorsiflexion myotome and a patellar reflex (L2, L3, L4)
 - L5: Diminished pain in the great toe and lateral calf, with ankle eversion myotomes and a diminished Achilles reflex (L5, S1)
- Sacroiliac (SI) joint symptoms are rarely above the L5 level; rather, they are along the spine midline or a few centimeters lateral. They are often

exacerbated by standing up and performing split lunges, and may be associated with a history of trauma. Typical wisdom is that generally three or more traditional SI joint provocation tests should be positive to conclude that SI joints are the pain source (but these do not isolate SI joint loading). We put more weight in the fist pulse test and the walking compression test described in the section Test for Sacroiliac Joint Pain. Interestingly, pain during provocation is on the unstable, hypermobile side, yet symptoms are more often on the hypomobile side. Clinicians often treat the wrong side!

- Facet joint pain can be radiating, and is usually slower in onset and in resolution. Pain is usually isolated with extension and rotation. Facet disorders are rarely the primary injury since they are usually secondary to discogenic disorders.
- Muscle pain is often mistaken as the only source given that it usually indicates underlying pathology. The pain usually peaks 24 to 48 hours following trigger exposure and responds to techniques such as active release therapy (ART) and contrast heat and cold. It is sometimes confused with neural tightness, particularly in the hamstrings.
- Spondylolisthesis is usually confirmed with radiological imaging, but symptom confirmation can be done with the spondylolisthesis test described later in this chapter. Evidence from MRI T2 images of fluid in facet interspaces is another strong indicator. People typically report having a continually “grumpy back” that is exacerbated by motion, prolonged postures, and sleeping postures. Standing extension pain is often lessened during the one-legged stork test.
- Stenosis may be confused with claudication (motor losses after stair climbing or walking). Symptoms have a slow onset and include pain (although sometimes not in the back) and radiating numbness. This is one of the few conditions that respond to traction and traction with extension. As with many of these syndromes, a key for recovery is to train in short, tolerable intervals.
- Compression fractures are most common in people who are osteoporotic, although people who participate in outdoor sports (hunting and fishing) also experienced them. Radiological confirmation (e.g., vertebral collapse, split vertebral end plates and sacrum, Schmorl’s nodes) is needed. The heel drop test conducted with care assists in confirming symptoms, which can range from local periosteum pain to

radiating symptoms.

- Ankylosing spondylitis has an insidious onset usually of morning stiffness and a progressing flexion antalgic posture. Lewit extension exercises performed daily may assist with posture and pain control.
- Arachnoiditis cases are some of the saddest I have seen. They are often the result of postsurgical nerve root scarring. This very debilitating condition may also be instigated by complications from epidurals, myelograms, and so on. We have had a few successes with nerve flossing and movement control.
- Psoas abscess is a rare condition usually associated with a history of fever, infection, and prior surgery. MRI and blood work are needed for confirmation.
- Fibromyalgia is usually characterized as swirling pain around the shoulders, hips, and back, with accompanying headache. Because more activity increases symptoms, exercise therapy can be challenging. I am convinced that people with fibromyalgia have usually been associated with a history of trauma sufficient to rewrite and rewire movement engrams. Thus, the brain perceives these movements as pain generators. The key, I believe, to the few successes that we have had is for people to find pain-free movement patterns and practice them. They should then expand their pain-free repertoires as their potential grows.
- Neurological engram disruption occasionally follows trauma. These patients have forgotten basic movement and motor patterns. They are unable to cross their legs when sitting, or patterns of rising from a chair are missing. This may occur as a result of motor cortex rewiring from a traumatic event or of traumatic spinal cord shock when the engrams that reside in the cord are lost. The therapeutic approach is to help people relearn movements with repeated, short-duration practice. Insist on impeccable form.
- Scheuermann's disease requires radiological confirmation, which shows wrinkled vertebral end plates. Patients report constant pain exacerbated by activity and workload. We have observed this to appear more frequently in taller and lankier males. If you coach the person in spine-sparing movements, he will likely grow out of these adolescent pains.
- Hip pain from the joint may radiate into the groin, scrotum, medial thigh, femur, and deep buttock. The hip is confirmed as the source with provocative testing isolating the hip capsule, labrum, and acetabulum.

Hip pain results in what is called gluteal amnesia and often psoas tension. This may be confused with piriformis syndrome, which is usually associated with blunt trauma to the buttock posterior hip joint region, or the sciatic nerve piercing the muscle (confirmed radiographically). Interestingly, hip and back pain often occur simultaneously because of the mechanical linkage. When hip extension or stiffness is different unilaterally, the spine often acquires a twisted rotation, which changes both static and dynamic mechanics.

- Complex symptoms may simply reflect a constellation of disorders. People with complex symptoms are sometimes labeled bizarre or, worse, accused of malingering or told that the pain is all in their heads. Proper interpretation is required to unravel these comorbid syndromes.

Assessing Posture and Movement Quality

Posture and movement patterns influence the loading of specific tissues. These habitual patterns result from engrams, which are encoded motor control patterns. Apart from acute trauma, these motor control patterns are probably the most important determinant of back injury. This opinion has been gained from both working as a consultant to athletic teams and programs and performing longitudinal studies. For example, I have consulted with a number of Olympic programs and sport organizations, both professional and amateur. Watching the movement patterns of individual athletes early in the season motivated me to write down who I thought would become injured and in what way. For one team, I predicted that two players would sustain back complaints that would affect their play (they tended to initiate movement with the spine rather than the hips, as their teammates did). I also predicted that two athletes would sustain ankle injuries because of the heavy pounding of poorly directed forces down their legs, and that one would sustain a knee injury because he used the knees to brake his motion rather than the hips. I correctly predicted four out of five injuries (the knee case did not materialize that season). Movement patterns predicted who would become injured. The corollary is that this knowledge could be used to intervene and reduce the risk.

Using a more scientifically rigorous example, we performed a study on 76 workers who performed the same physical job (McGill et al., 2003). A total of 26 had recurrent back episodes lasting on average a couple of weeks per year, but all were at work and functioning pain free on the testing day. Those who had recurrent painful back attacks differed from their healthy colleagues. When bending to pick up a coin from the floor, they bent their spines more and their hips less than their healthy colleagues did. They had more back strength when tested. We assumed that this was because they overused their backs when working. They also had less muscle endurance. It became clear that although they were stronger, their lack of endurance caused them to break form when lifting, imposing stresses to their back that led to constant tissue irritation. They had stiffer hips, particularly in flexion and internal rotation. Interestingly, psychosocial markers were present in the recurrent pain workers, on average, but they were less important than those considered biomechanical and motor control in nature.

In a study of adolescents, Smith, O'Sullivan, and Straker (2008) showed higher odds for having back pain with nonneutral sagittal spine postures when standing. It's interesting that skilled clinicians can often pinpoint the location of the pain index simply by seeing the region of highest deviation or the location of a spinal hinge observed in bending or during the cat–camel exercise.

The combination of biomechanics and motor control variables in provocative tests and movement assessment screens is perhaps the most powerful component of an examination of a person with back pain. However, just because a person can move in a certain way during testing does not mean that she will move in a joint-sparing way when performing tasks of daily living. This means that a simple movement screen, one that is performed quasi-statically and without substantial external load, does not predict how a person will move in other tasks. We studied the movement patterns of firefighters in simple tests and then quantified their movements while they performed tasks such as carrying ladders, chopping holes in roofs, and pulling hoses. The simple tests did not predict how they loaded their bodies when loads and speeds were increased to real-life conditions (Frost et al., in press). In another study, we screened 180 students to quantify their squat and lunge mechanics together with their hip and back range of motion. Interestingly, there appeared to be little relationship between whether they could move in a joint-sparing way versus whether they chose to move in a joint-sparing way. In other words, just because a person can move in a certain way in a controlled test doesn't mean he will move that way in real life. Motor control patterns, or patterns of movement, need to be trained in certain people. Similarly, people should be assessed while performing real-life tasks.

Our movement assessment begins during the interview by observing the person sit. Then we observe sitting-to-standing, walking, and pulling to open doors while walking from the interview office to the assessment clinic. We look for hip-hinge patterns versus spine bending, walking with stiff hips and shoulders, pulling with spine motion rather than hip movement, and a stiffened spine. These patterns were introduced in chapter 8.

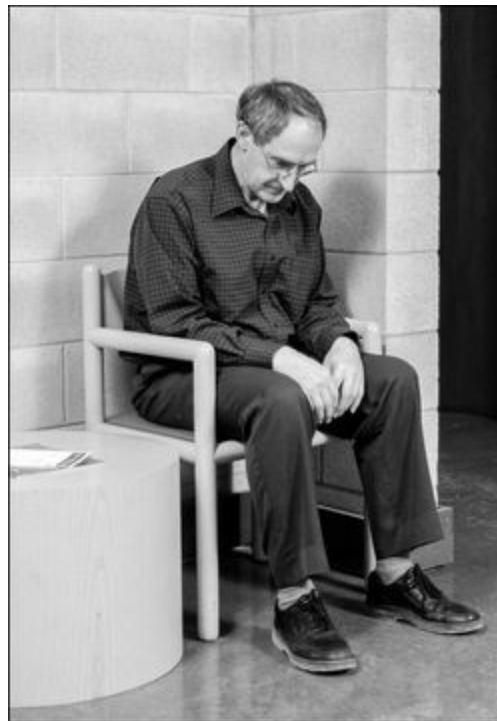
Formal Assessment of Posture

A formal assessment of posture includes an assessment of sitting,

standing, and lying. Pain is present because the back is currently a weak link. Functional tests help to sort out why the back is a weak link and help indicate what is required to bolster the deficit. Some tests are posturally based, whereas others assess control of motion. The following postural tests are very helpful. Recall that screening need not involve manipulation or special positions. Some of the most useful “tests” are simply intelligent observation.

Sitting Posture

Remarkably, patients with troubled backs often sit in a way that causes more back troubles. For example, patients who show flexion intolerance during provocative tests often have a typical sitting posture that involves a lot of spine flexion. In a similar way, those who show extension intolerance often sit with the spine locked into extension. The sitting posture is the first clue in hypothesis formation because this is often the posture that is first observed in the waiting area. Sitting in the waiting room in a flexed posture often suggests flexion intolerance.



Photos from Stuart McGill

After noting the sitting posture, observe how the patient shifts from a slouched sitting posture to an upright one. Did she lift the rib cage, stressing

the thoracolumbar junction (a and b). Or did she flex the hips, rolling the pelvis forward to align the spine (c). Our recent work has shown that the best choice for most patients is to use a combination of the two (Castanhero et al., 2014) to minimize stress and avoid pain.

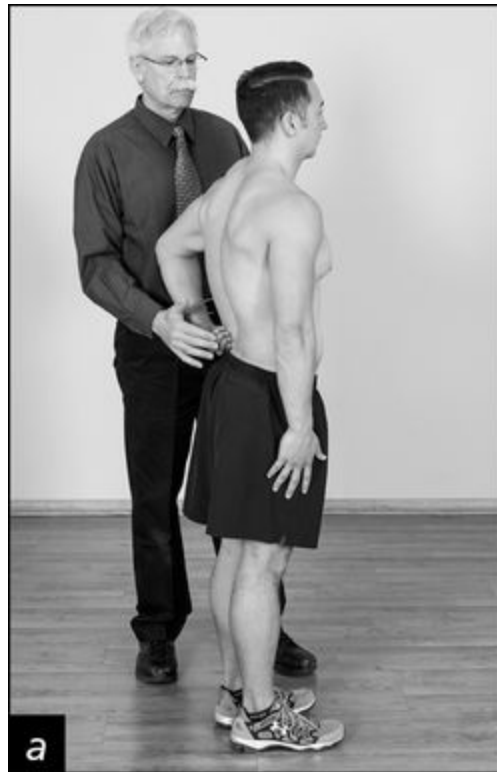


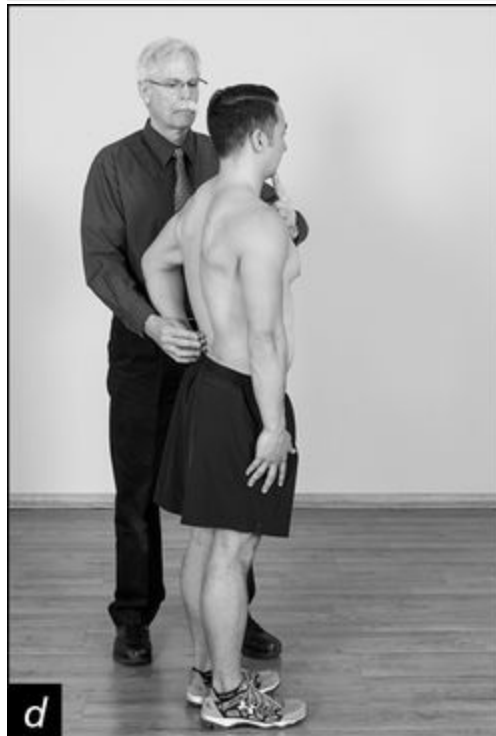
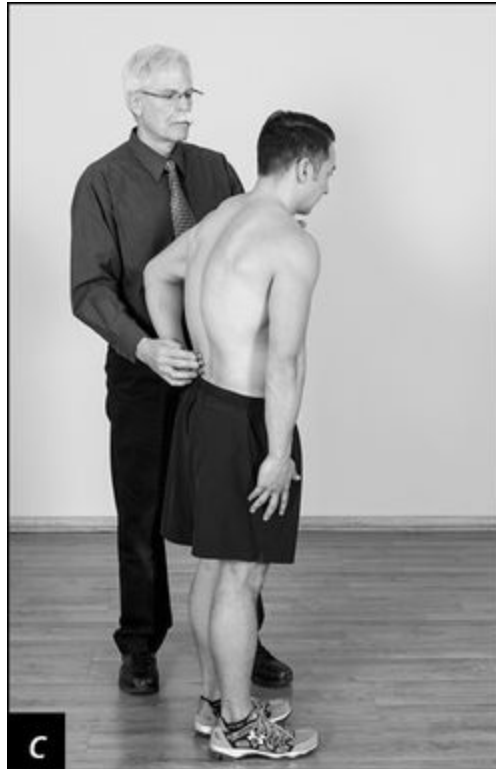
Does the patient sit with a neutral spine (a) or a flexed spine (b)? When shifting from a slouch to a more upright posture, is the movement concentrated about the thoracolumbar junction or hips (c)?

Standing Pattern

All sorts of clues are revealed in the standing posture. Simple palpation of

the standing patient's lumbar extensor muscles will reveal whether the patient is chronically crushing the back with extensor contraction. Simple postural corrections such as extending the hip, retracting the shoulders posteriorly, and pulling in the chin can shut these muscles off.

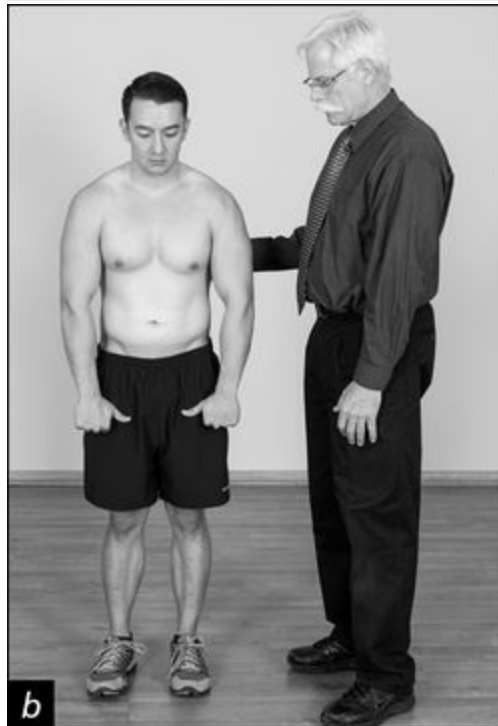
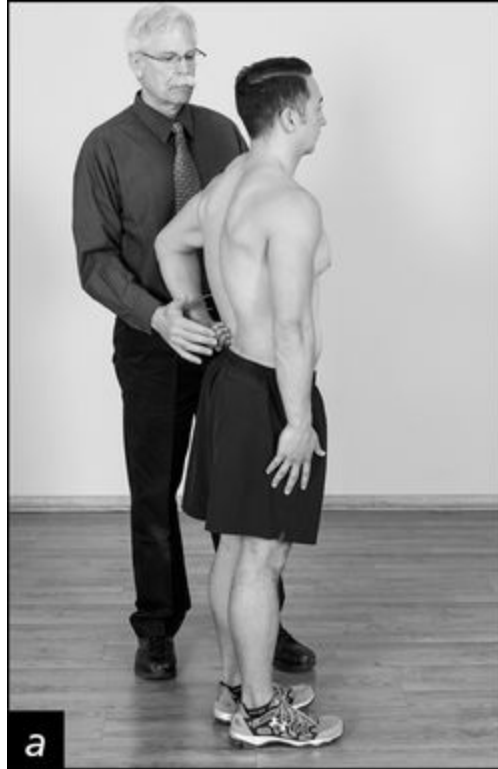


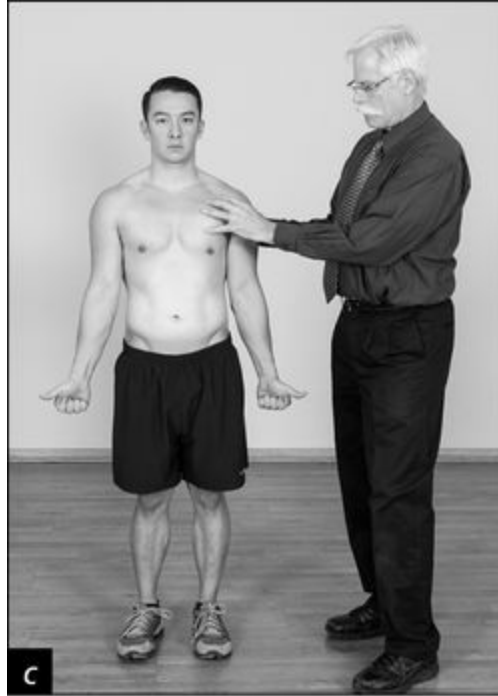


Teach patients to palpate the erectors at the L3 level, and learn where the switch point is between muscle activation and muscle relaxation. They do this by standing upright, head erect and shoulders back, and then leaning

forward slightly (a). Then they perfect this strategy to relax the muscle. Teach them that chin poking (b) and shoulder slouching (c) activate the muscles that cause fatigue cramps. Have them try the hitchhiking thumb position as a correction strategy, externally rotating the arms about the shoulders and then letting the shoulders relax into this new position (d). Having patients palpate the lumbar erectors helps them learn self-correcting strategies—are the erectors silent or active (a)? If they are active, show the patient how to shut them off by unloading the spine, which often provides immediate relief of muscular pain. Cues include shifting the head and neck from a chin poke to chin retraction position, adopting a hitchhiking posture with external rotation of the arms and shoulders, and pulling the hips forward (b, c).







No amount of muscle relaxants will shut off the painful back muscles if the patient stands in a slouched posture. Standing to spare the spine requires a finely tuned blend of postural adjustments and muscle activation patterns. One correction is to steer the thumbs in external rotation, retracting the shoulders (see b and c) to a spine-sparing, corrected posture.

Some patients stand with no measurable extensor or abdominal muscle activation—they simply balance on their passive elastic spine tissues. These people tend to have thick torsos with flaccid abdominal walls. For these patients, pain reduction can be achieved with mild abdominal contraction; 2 or 3% of maximal voluntary contraction is considered normal abdominal wall muscle tone during standing (Juker et al., 1998).

Lying Pattern

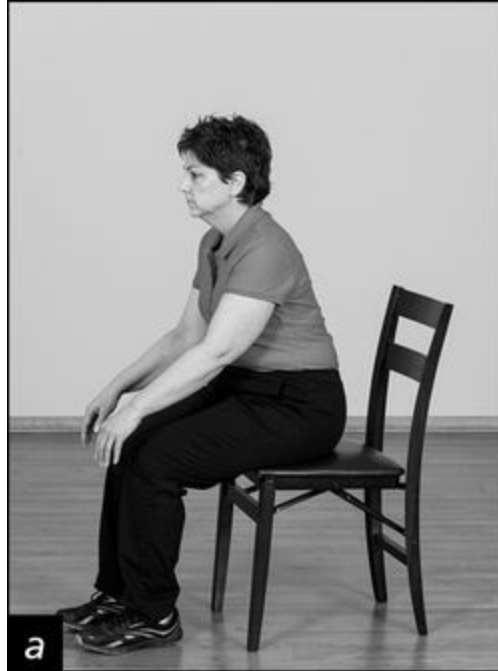
Stress during lying is a function of body shape, the lying surface, and the pain mechanism. For back sleepers with protruding buttocks, lying on a hard surface causes a bridge under the lumbar spine. This causes spine flexion as the back flattens to the mattress and pain in the flexion-intolerant back. This may be addressed with support such as a folded towel (a, b). In a similar fashion, wide hips cause full, and painful, spine lateral bend in side-lying. In this case, a pillow-top surface can provide relief.

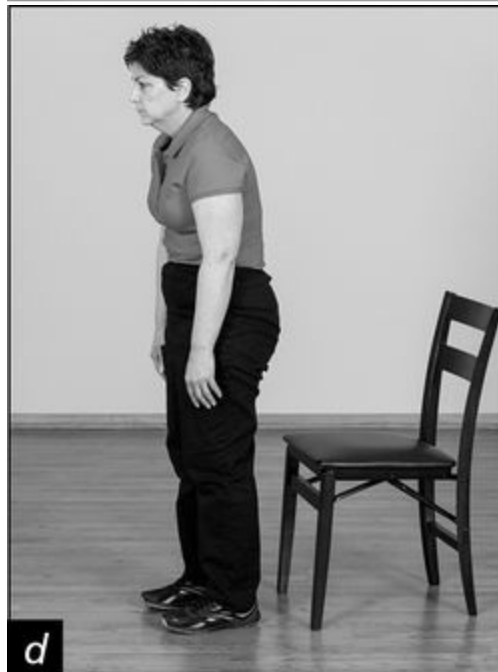


(a-b) Painful backs with lying and sleeping postures are usually relieved with support to foster a neutral posture. This support may be in the form of mattress selection (adjusting firmness and the thickness of the pillow-top) or buttressing devices such as a folded towel in a person whose spine is bridged with protruding buttocks (a). A pillow to support the neck and knees may assist with side lying (b).

Formal Assessment of Movement

Qualitative observations begin the instant the patient is greeted. At our clinic the assessment room is downstairs from the interview room, which provides an opportunity for observing sitting-to-standing, walking, descending stairs, and pulling and pushing heavy doors. A more formal assessment of basic movement patterns is conducted following the interview. A poor and painful pattern (a-d) and a better spine conserving pattern (e-h) is shown.









Rising From a Chair

Rising from a chair can be very revealing of numerous motion and motor patterns. For example, the flexion-intolerant patient initiates rising from a chair first by lumbar flexion and even sometimes hip extension. Such patients' movements exacerbate the painful tissues, preventing recovery. A

conserving approach for the flexion-intolerant person would be to initiate the movement with a lifting of the chest, causing lumbar extension first with corresponding hip flexion motion. The feet are underneath the center of mass of the body, and hip extension drive is dominated by gluteal muscle activation. Hip power in this case is often helped by shifting the pelvis forward in the chair and planting the feet wider apart on the floor.



Also, rising from a chair often reveals spine hinges. These are regions, or motion segments, in which a disproportionate amount of the motion occurs. These overused regions are often the site of local symptoms, and until they are addressed, the back often remains chronic.

Assessing Basic Patterns of Movement

Motion restrictions other than in the spine may also be observed when a patient rises from a chair—for example in the hip, knee, or ankle. These restrictions may be complicating the diagnostic hypothesis and may or may not become a component of the back therapy.

Squat: Observe where movement is initiated—is it in the spine, hips, knees? Teach the shortstop squat position, with appropriate corrections. Then

drop a coin on the floor and ask the person to pick it up. Observe the choice of movement strategy. If the person squats, point out that this is the wrong movement tool. Demonstrate the golfer's lift.

Lift: Have the person lift a heavy object (appropriately heavy for them). Teach appropriate torso stiffness, the antishrug, the hip hinge, and appropriate bracing for movement and pain control.

Lunge: Observe the basic lunge pattern and correct to help the person maintain an upright torso. Then have him lie on the floor. Did he choose the lunge with a crawl pattern as the preferred movement tool? (See chapter 8.)

Pull: Have the patient open a door (a). Did he lean forward with the torso, and did he stiffen the torso? (b)





Here the patient flexes the spine to reach for a door (a). Because the spine has little stiffness, the shear forces create pain. Preparatory muscle bracing is absent (b). Correct these flaws by teaching the patient to prepare to apply load with bracing and to maintain the brace until the door is released and the body is upright.

Push: Assess the strategy in the same way as with the push.

Gait Pattern

Is walking a provoking task for the patient? If so, can it be turned into therapy? Many clues can be obtained during the observation of walking patterns. For example, a slouched posture, small steps, swinging arms from the elbows, duck feet, and a low-cadence shuffle are all suggestions that walking is an exacerbating activity (a). Correcting these will turn walking from an exacerbating activity into therapy. Lifting the chest, swinging the arms from the shoulders, taking longer and faster steps, or correcting duck feet often relieves the pain (b). The biomechanical changes show more efficient storage and recovery of elastic energy, actually reducing spine loads with this approach. Further, walking is changed from a static spine activity into a reciprocating motion and load activity.



Observing the dynamics of the rib cage relative to the pelvis is also revealing. A lumbar spine may be out of control with large motion; pain in these patients is often relieved with abdominal bracing patterns. Yet others with lumbar symptoms may be overly stiff in the thoracic and shoulder regions, with motion occurring only in the lumbar region. Too many young males who are bodybuilding fall into this category. The correction is to have them walk with little stiffness above the lumbar region and with some stiffness in the lumbar region. Developing an attitude during walking often

helps. Using cues such as “Walk like you own the world” or “Walk with no worries” often lightens the upper regions to unload the painful lumbar spine.